

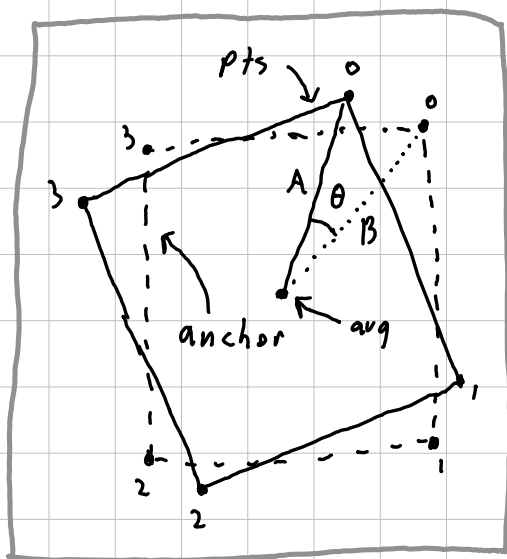
Shape Matching

→ rewatch walaber

anchor_pos → avg

anchor_rot

anchor_points → const



* anchor is in local space

recall

$$A \times B = |A||B| \sin \theta \leftarrow y$$

$$A \cdot B = |A||B| \cos \theta \leftarrow x$$

$$\theta = \text{atan2}(y, x)$$

for each point

$$A = \text{pts}[i] - \text{ctr}, B = \text{anchor}[i]$$

$$\text{dot} = \frac{A_x B_x + A_y B_y}{|A||B|}, \text{cross} = \frac{A_x B_y - A_y B_x}{|A||B|}$$

average angles?

$$x\text{-sum} = 0, y\text{-sum} = 0$$

$$x\text{-sum} += \text{dot}, y\text{-sum} += \text{cross}$$

then find theta after

$$\theta = \text{atan2}(y\text{-sum}, x\text{-sum})$$

note

"Semi-Static" shapes can have animations on rot, pos

i.e. gears on moving platforms

